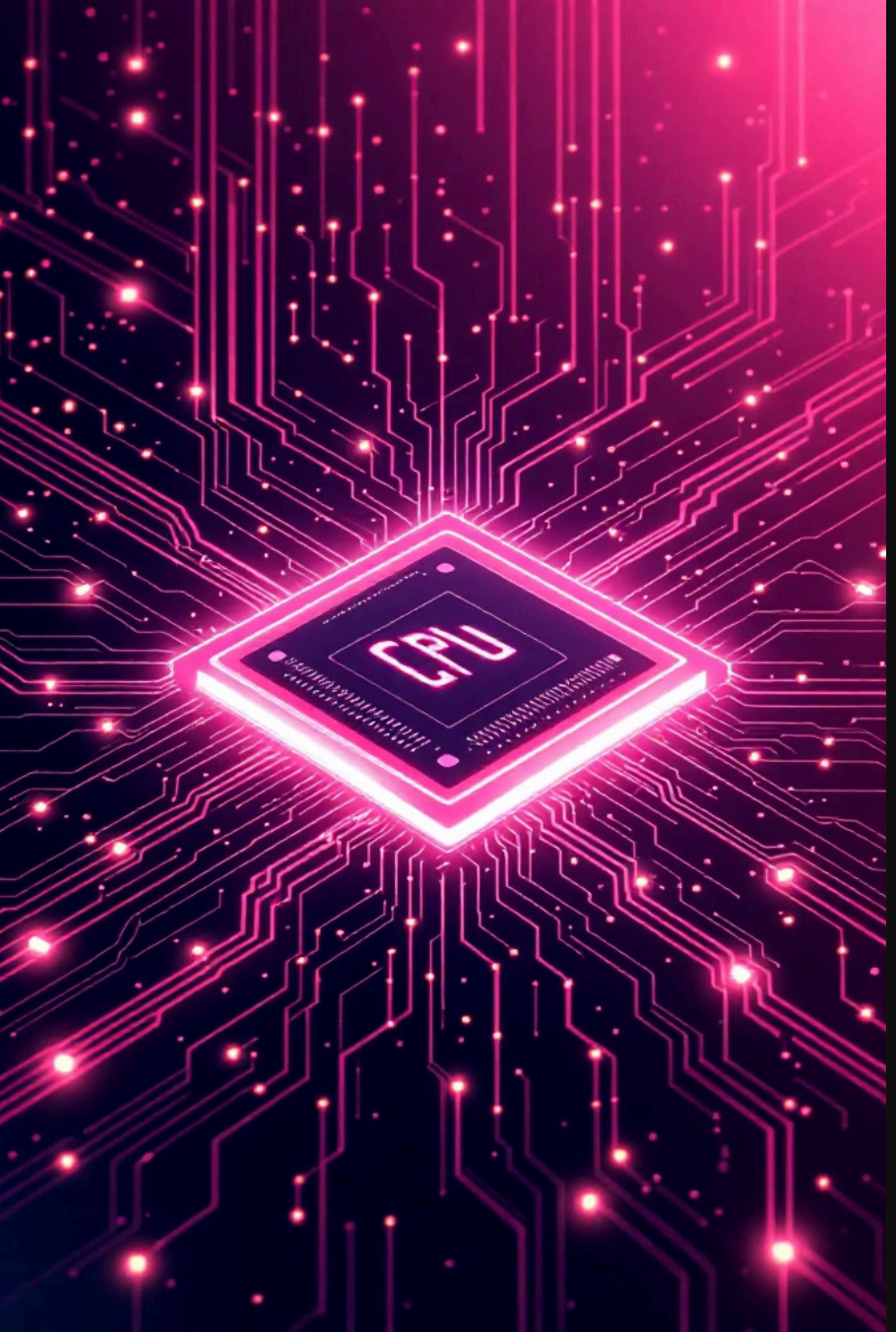


# The Intelligent System: CPU & Memory Analysis with ML Prediction

By Madhumitha M

Made with GAMMA

# From Reactive to Predictive: The Evolution of System Analysis

1

### Early Systems

Static thresholds, manual checks for CPU and memory

2

### Reactive Approach

False positives, missed critical issues

3

### Proactive Future

Intelligent monitoring with ML prediction

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# The Dawn of AI in Infrastructure Monitoring

### Traditional Monitoring

Alerting when CPU > 90% or memory < 10%

### Bleemeo's AI

Learns normal behavior, detects anomalies with dynamic bounds

### Predictive Planning

Linear regression forecasts disk full dates days/weeks ahead

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# Scalene: High-Resolution Profiling with AI Suggestions

## Low-Overhead Profiling

CPU, GPU, and memory analysis for Python

## AI-Powered Insights

LLM-based optimization suggestions

## Actionable Advice

Identify inefficiencies instantly

## Latest Release

v2.2.1 pushes performance boundaries

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# Deep Dive into Resource Utilization: Azure ML

## Key Features

- Profile models for CPU and memory needs
- Deploy as web services with adequate resources
- Generate recommendations based on observed usage

**Important:** SDK v1 deprecated. Support ends June 30, 2026

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# xMem: Predicting GPU Memory with CPU-Based Analysis

**91%**

**Error Reduction**

Decrease in median relative error

**75%**

**OOM Prevention**

Reduction in out-of-memory probability

Uses CPU-only dynamic analysis to accurately estimate peak GPU memory requirements in shared environments

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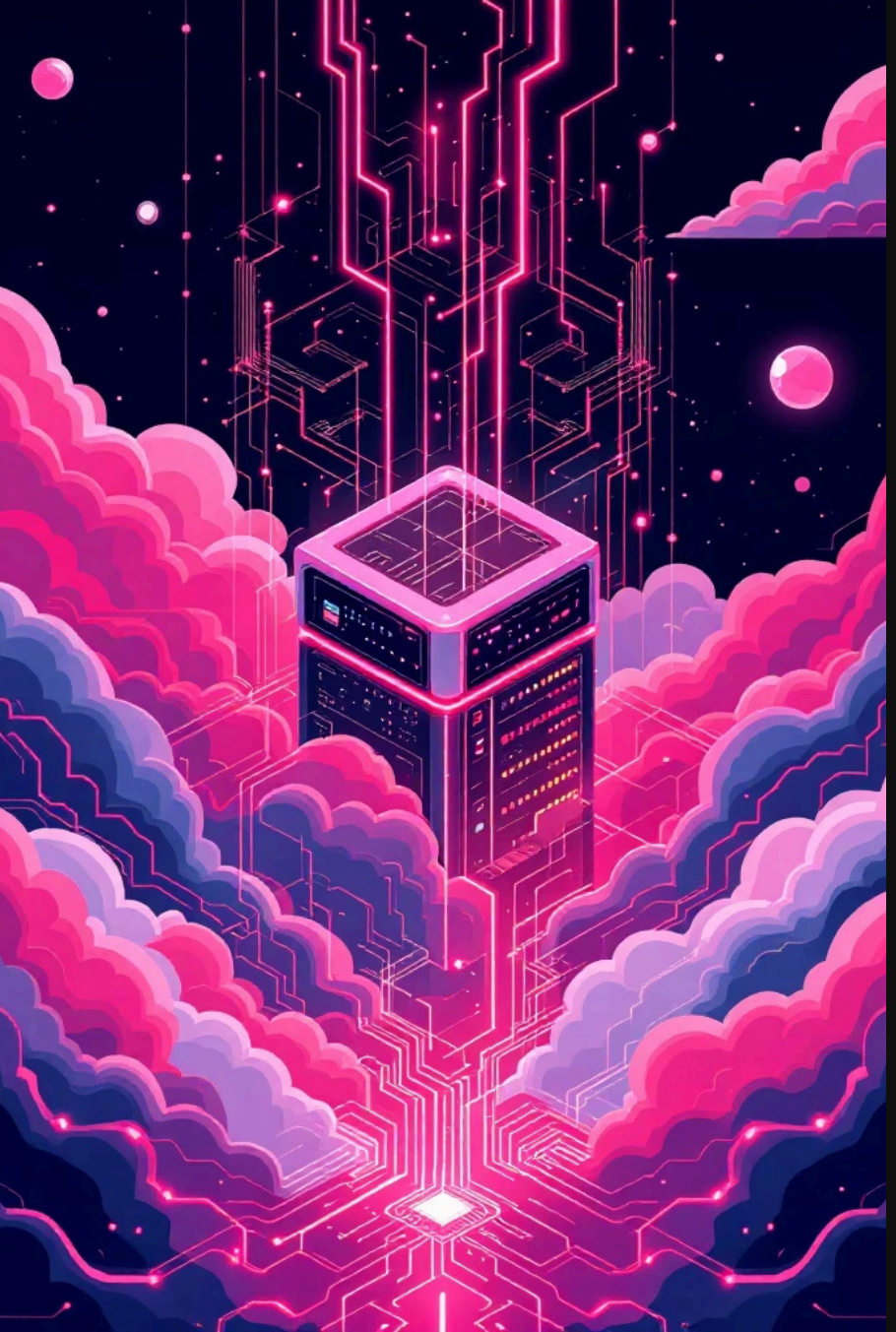

# Elasticsearch ML: Estimating Anomaly Detection Job Memory

01

## Elasticsearch ML Features

APIs to estimate model memory usage

02

## Estimate Model Memory API

Analyzes configuration details and field cardinality

03

## Resource Optimization

Crucial for anomaly detection job efficiency

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# System Analysis in Rust: Comprehensive Workload Modeling

### Hardware Assessment

Detailed analysis using system-analysis Rust library

### AI/ML Workloads

Compatibility checking and resource prediction

### Bottleneck Identification

Suggests optimal hardware configurations

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# The Future: Proactive, Intelligent Resource Management

### AI & ML Transformation

Reactive to predictive analysis

### Advanced Tools

Scalene, xMem, Bleemeo insights

### Accelerated Development

Faster innovation cycles

### Efficient Utilization

Prevents costly outages

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# Empowering Developers: Smarter Systems, Faster Innovation

### Robust Systems

Understand and predict resource needs

### Code Optimization

AI-driven analysis empowers developers

### Self-Optimizing

Journey towards intelligent systems underway

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